

11 Effects of Stretch Training on Muscle Strength and Hypertrophy

Konstantin Warneke

Chronic stretch training is considered one of the most common training interventions to enhance the range of motion (ROM). A plethora of research has been conducted over the years to investigate the effects of different stretching interventions, concluding that static stretching was one of the most effective techniques. While previous chapters provided definitions of different stretching methods that are used in human research, a more physiological point of view might “stretch” our perspective of this research field. When opening a new scientific field of potential interest, it is very common to start investigations using animal models (7). Before diving into a recently emerged field of human research, we will introduce the history of stretching research by briefly providing early evidence from studies with birds, cats, and rabbits.

What humans can learn from chicken and quails

The foundation of probably most of the known stretching research was established as early as 1887, using animal models. Marey (10) removed the attachment of the soleus muscle and attached it farther away from the skeletal origin. While this surgery might not be considered a relevant training intervention, the muscle was chronically stretched and responded with a rapid enhancement in the number of sarcomeres in series, (sarcomerogenesis), which resulted in an enhanced muscle length. However, to induce a chronic mechanical overload, it is not necessary to invasively move the muscle attachment. I am sure you can imagine what the human ethics committees might say if we tried these experiments on our students or the public. A series of investigations starting in the second half of the 20th century applied chronic stretches to the wing muscles of chickens and quails by attaching weights or stretching apparatus that abducted the adductors. By adopting very different protocols, using stretching durations that ranged from 30 minutes per day for five weeks to chronic interventions of 24 hours per day, seven days per week for up to six weeks, the various researchers consistently showed large muscle mass increases in the stretched muscle (18, 19, 20, 21, 22), which followed a duration-response relationship (more stretching = more hypertrophy). However, the enhanced muscle mass was not exclusively attributable to

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serial sarcomere attachments, or in other words, increases in muscle length. As early as after the first day of stretching overload, the authors reported additional muscle and fibre cross-sectional increases, while chronic training protocols resulted in a significantly increased number of muscle fibres (hyperplasia). These increases were accompanied by a significant increase in muscle protein DNA and RNA resulting in a stimulated net protein synthesis rate, resulting in an increase in total amount of protein, RNA and DNA by 59 percent, 228 percent, and 82 percent respectively after seven days (17) that were attributed to mechanical tissue overload from passive stretching. However, these results were questioned, as it could not be ruled out that attaching a weight or stretching apparatus to the birds would initiate active muscle contractions. To explore the role of active contractions, Sola, Christensen, and Martin (16) removed the innervating nerve for the stretched muscle and compared the adaptations with those of innervated wing muscles. In both groups, there were substantial muscle mass increases with a slightly smaller effect in the denervated muscle. Accordingly, in 1980, Barnett and colleagues (1) and Holly et al. (6) confirmed no electromyography increases in the chronically stretched muscle, underlining the potential of stretch-induced mechanical overload. Thus, even prolonged passive stretching with its tension on the tissues that are typically lower than experienced with resistance training can induce muscle hypertrophy. For more detailed underlying mechanisms and the results of individual studies can be reviewed in Warneke et al. (23). Briefly, the tension on muscle whether it is from resistance training or static stretching will trigger the integrin molecules on the muscle membranes to initiate a cascade of events that initiate protein synthesis through the Raptor/mTOR complex. I find an easy way to remember the function of the integrin molecule is to remember that it monitors the "INTEGRITY" of the membrane and thus if the integrity is being threatened by tension, it triggers protein synthesis to protect it. While active contractions will add a number of other contributing mechanisms, passive tension with stretching alone can still promote muscle hypertrophy.

Interestingly, Frankeny et al. (5) and Bates (2) showed that chronic (24 hours/7 days per week) stretching interventions in animal models were unnecessary when increasing muscle strength and mass. Both studies used intermittent stretching protocols and confirmed significant muscle mass increases after at least 30 minutes of stretch. Nonetheless, longer stretching durations might be not economical. Bates (2) allocated 30 chickens into different stretching groups that performed daily anterior latissimus dorsi stretching for five weeks. While 30 minutes daily stretching resulted in 57 percent mass increase, doubling the stretching time provided only additional 2 percent. Another doubling of the stretching time (two hours of daily stretching) resulted in a 67 percent increase, while an additional two hours showed 72 percent muscle mass increase after four hours of daily stretching. Also Frankeny, Holly and Ashmore (5: pp. 275–276) summarized that "Thirty minutes of stretching per day is certainly within normal physiological limits, and as a result may be applied to human muscle with hopes that similar adaptations

would occur”. The authors suggested exploring the transferability of animal results to humans. Two reasons might explain the lack of longer duration stretch training studies in humans until 2021. The first reason associated with animal data collection procedures is obvious. All animal participants are terminated after the prescribed intervention period, with the muscle removed and assessed via light microscopy. Indubitably, recruiting human volunteers for such a study might be a little difficult! ☹ Secondly, it is difficult to recruit enough participants who are willing to spend up to several hours per day stretching, in addition to their other activities of daily life.

Does stretching provide a sufficient stimulus to enhance human muscle strength and size?

These experimental burdens prevented designing studies that explored the transferability of these animal stretch results to humans. While an early review article suggested a possible benefit of stretch training for performance parameters such as jumping, sprinting, or strength-related performance (14), human research was rare. Kokkonen and colleagues (9) stretched lower extremity muscles with 3x15 seconds per muscle for an overall training time of 40 minutes per session, three days per week for ten weeks and demonstrated significant increases of 32.4 percent in the knee extensors, 15.3 percent in the knee flexors, with improvements in sprinting and jumping performance. Recruiting female track and field athletes, Bazett-Jones et al. (3) did not confirm the benefits of stretching on athletic performance. Within the following years, other researchers conducted a variety of static stretching interventions on muscle strength and hypertrophy, resulting in inconclusive results with studies showing stretch-induced strength and muscle size improvements, while others did not. These controversial results might arise from a high heterogeneity in the stretching methods that prevented Medeiros and Lima (11) from performing a meta-analysis. While in 2017 Medeiros and Lima (11) reported a potential benefit of stretching on performance, in 2020, Nunes et al. (12) reviewed the existing literature without confirming stretch-mediated hypertrophy in humans. However, the authors highlighted that a lack of research performed high intensity/high volume stretching interventions. Indeed, it would be inconsistent to compare five minutes or less of continuous stretching per session in humans, with animal research that implemented a minimum of 30 minutes daily stretching.

Interestingly, in 2021 researchers from Japan and Greece initiated stretching studies using a stretching board for 6x5 minutes, two days per week or up to 5x15 minutes for five days per week (13). While in the Yahata (27) study, strength increases of about 10 percent were not accompanied by significant muscle hypertrophy, Panidi and colleagues (13) reported large effect size magnitudes (24 percent) of muscle hypertrophy. Both studies suggested the applied stretching dosage to be of crucial importance for stretching responses regarding strength and hypertrophy.

In 2022 the first studies using training loads more similar to animal research were conducted (18, 19, 20, 21, 22). Applying stretching partly via a stretching orthosis, these works performed stretching for 5–120 minutes per session, 6–7 days per week for six weeks. By using ultrasound- and magnetic resonance imaging (MRI), significant stretch-induced muscle hypertrophy in the plantar flexors (16 percent), that were accompanied by maximal strength increases of up to 25 percent were confirmed.

Underlying mechanisms of stretch-mediated hypertrophy and strength increases

In contrast to animal research that showed an improved muscle protein synthesis that was attributed to mechanical overload-induced stimulation of anabolic signalling pathways (23), the underlying physiology in humans is rarely investigated. In 1993 a study by Smith and colleagues (15) described potential indications of mechanical overuse induced with stretching by finding enhanced creatine kinase values. However, these results could not be confirmed by Wohllann et al. (25, 26) Moreover, the only stretching study that assessed stretch-induced protein synthesis changes was not able to find significant changes (4). However, as these studies did not use appropriate stretching volumes (e.g., insufficient durations with an overall volume of 33 minutes), the role of stretching-induced mechanical overload and subsequent enhanced protein synthesis remains speculative in humans.

Pointing in another direction, Hotta et al. (8) stretched rat hindlimbs for 30 minutes per day for four weeks and found blood flow restrictions while stretching, which caused an enhanced capillarization. Also in humans, the effects of restricted blood flow on muscle hypertrophy are well investigated. Even though the literature lacks evidence that reports decreased blood flow while performing 30–120 minutes of stretching per day in humans, it seems a concurrent, but logical alternative explanation.

Consequently, current systematic reviews report significant, small magnitude improvements in maximal strength and muscle size with static stretching, if performed with sufficient volume, intensity, and frequency. However, further research is required to understand the underlying physiological processes to maximize and optimize these effects.

Practical applications and limitations

Although it is possible to enhance muscle strength and size without performing active contractions to a similar magnitude as with commonly performed resistance training in the plantar flexors and pectoralis muscle, one might argue that there is limited practical relevance, owing to the comparatively long durations required with static stretching. To reach comparable results in the calf muscles, seven hours of stretching, four days per week were necessary, while Wohllann and colleagues (25, 26) used a stretching apparatus and

performed 15 minutes of uncomfortable stretching including constant re-adjustment (increasing) of the stretching tension to counteract creeping and relaxation effects. Especially as these training volumes were applied to just one muscle, the exercise routine seems of limited economy (time restrictions) compared with active resistance training, that may use 20 percent of stretch training durations to reach almost similar results. Nevertheless, there might be some populations with restricted accessibility to training equipment, or after serious injuries that cause immobilization of a limb, for whom active resistance training might be contraindicated. For those individuals, but also the unmotivated populations, high volume of passive stretching by using a stretching device that enables the integration of this workout into the daily seated times while watching television or working in the office could provide a valuable supplementation. Active movements however, if applicable might provide further benefits such as improved cardiovascular health, neuromuscular adaptations, or transfer effects to athletic performance parameters such as sprinting or jumping. Even though some high-volume stretching studies showed first indications of enhanced squat jump performance by stretching, Warneke et al. (24) were not able to replicate the benefits of chronic stretching routines on jumping or sprinting performance. Thus, using active exercise routines would be preferred when aiming to enhance strength and hypertrophy, when applicable, but there are scenarios where stretch-induced muscle strength and hypertrophy may be a suitable alternative.

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